

Layer Stack Legend

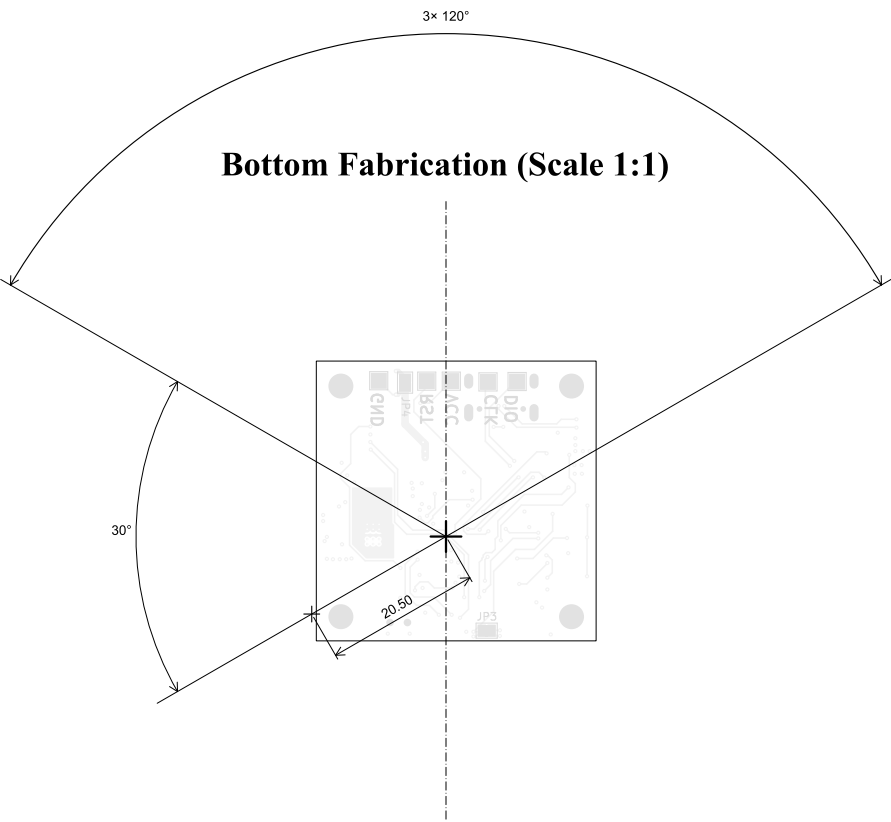
Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

12) CONFIRM SPACE WIDTHS AND SPACINGS.

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FCU3 Fabrication Document



All dimensions are in millimeters unless otherwise specified.

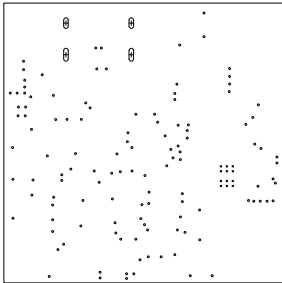
	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4c2c47c
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 10

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	12	0,20mm (7,87mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
O	121	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
Total 137						

Drill Drawing L1 - L4 (Scale 1:1)



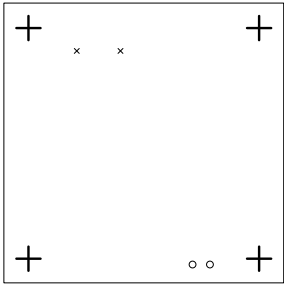
	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4c2c47c
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 10

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	2	0,85mm (25,59mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
O	2	0,90mm (35,43mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
+	4	3,20mm (125,98mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
Total 8						

Drill Drawing L1 - L4 (Scale 1:1)



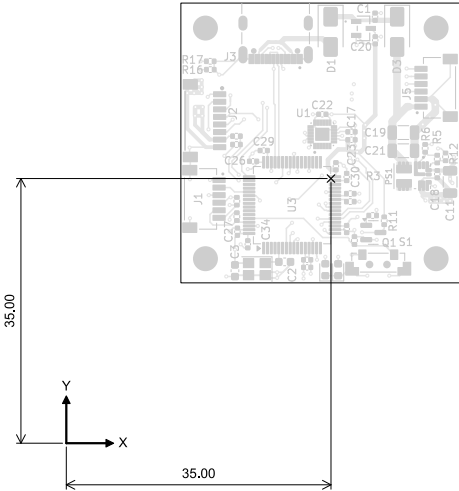
	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4c2c47c
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 10

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Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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Ref.	Net	X [mm]	Y [mm]
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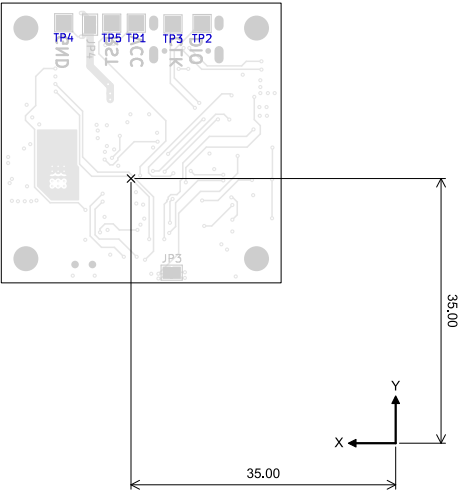
All dimensions are in millimeters unless otherwise specified.

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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Top Test Points (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 10

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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	+3V3	83.80	36.55
TP2	SWDIO	75.10	36.50
TP3	SWCLK	78.95	36.45
TP4	DGND	93.40	36.60
TP5	Net-(U3-NRST)	87.05	36.55

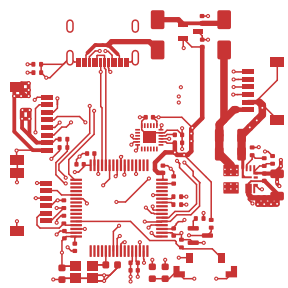


All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4c2c47c
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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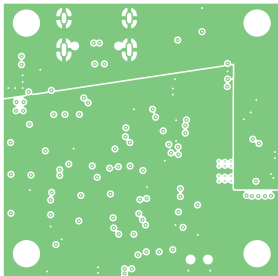
L1 (Sig, PWR) (Scale 1:1)



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		Kaldstrand Electronics		CHECKED	4c2c47c
	Sheet Title:	Board Name:		Project Name:	
		FCU3		Flight Controller 3	
	L1 (Sig, PWR) (Scale 1:1)	File Name:	Designer:	Date:	Revision:
	Sheet Path:	Reviewer:	Size:	Sheet:	
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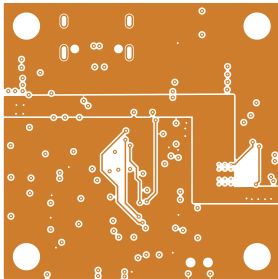
L2 (GND) (Scale 1:1)



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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L2 (GND) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 8 of 10

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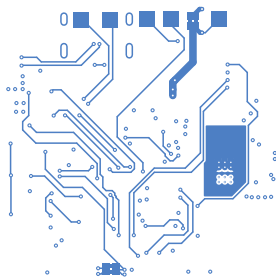
L3 (Sig, PWR) (Scale 1:1)



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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L3 (Sig, PWR) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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L4 (Sig, PWR) (Scale 1:1)



	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4c2c47c
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 10 of 10